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DATA STRUCTURES AND ALGORITHMS - 3 (25CS2103E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-13	Team No-01	Project Title-Plagiarism Detection Engine for Academic Submissions

1. Abstract

The **Plagiarism Detection Engine for Academic Submissions** is a **Data Structures and Algorithms (DSA)-based project** developed to identify similar or potentially copied content among academic documents such as assignments, reports, and student submissions. With the increasing number of digital submissions in educational institutions, manually comparing every document becomes time-consuming, repetitive, and difficult to manage. This project provides an efficient computational approach by applying appropriate data structures and algorithms to organize, process, search, compare, and analyze textual content.

The system accepts multiple academic submissions and processes their text by dividing the content into words or tokens. **Arrays or Python Lists** are used to store and manage multiple documents and their processed words. **Hash-based data structures, such as Python Dictionaries/HashMaps**, are used to store word frequencies and enable efficient retrieval and comparison of document content. **Searching techniques** are applied to locate specific words or matching content, while **sorting algorithms** can organize plagiarism results based on their similarity percentages. Additional data structures such as **Stacks and Queues** may be used to manage comparison history and process multiple submissions systematically.

The plagiarism detection process compares the content of two or more submissions, identifies common or matching words, and calculates a similarity percentage based on the amount of matching content. The generated results help identify documents with a higher possibility of plagiarism and allow users to prioritize them for further review. The project primarily demonstrates the practical implementation of fundamental DSA concepts rather than relying on advanced Artificial Intelligence or Machine Learning techniques.

Overall, the **Plagiarism Detection Engine for Academic Submissions** shows how data structures such as **Lists, HashMaps, Stacks, and Queues**, along with algorithms for **searching, sorting, and string comparison**, can be combined to solve a real-world academic problem. The system aims to reduce manual effort, improve the organization of submissions, and provide a basic, efficient, and understandable approach to detecting textual similarity. In the future, the project can be enhanced with advanced string-matching algorithms, phrase-level comparison, support for multiple file formats, and larger-scale document processing.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520080015	V. Hari Priya	
2520080058	G. Anirudh	
2520080068	M. Sai Harika	

3. Frameworks / Platforms / Software Used

Programming Language

- Python 3.x
  Used as the primary programming language to implement the plagiarism detection logic and DSA concepts.

Platform / Development Environment

- Visual Studio Code (VS Code)
  Used for writing, executing, debugging, and testing the Python program.

Python Data Structures

- List – Used to store multiple academic submissions and words.
- Dictionary (HashMap) – Used to store word frequencies and perform efficient lookups.
- Set – Used to identify unique and common words between documents.
- Stack – Can be used to maintain comparison history or undo operations.
- Queue / Deque – Used to process multiple document submissions in sequence.

Algorithms Used

- Linear Search – Used to search for specific words or matching content.
- Binary Search – Can be used for efficient searching in sorted data.
- Sorting Algorithms – Used to arrange plagiarism results based on similarity percentage.
- String Comparison Algorithm – Used to identify matching words or content between documents.

Python Libraries / Modules

- collections – Used for data structures such as `deque` and `Counter`.
- re (Regular Expressions) – Used for basic text cleaning and processing.
- os – Used for handling and accessing files.

Operating System

- Windows 10 / Windows 11
  Used as the platform for developing and running the project.

Version Control

- Git and GitHub
  Can be used to manage source code versions and project development.

Guide Name and Signature (remarks if any) _____